

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ae555280a8f0824c4.jsonl`  
- SHA-256: `37253f31e88b9424e0c5c0497ee23cd1c75dfcaef729e559769d1b2aa994dfa6`  
- Source modified: `2026-05-31T04:24:04+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:23:17.949Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"olmOCR-2-7B is a document-OCR-specialist VLM fine-tuned from Qwen2.5-VL-7B-Instruct, with additional GRPO RL training specifically to improve performance on math equations and tables ? the harder cases for bank-statement parsing."

****Source:**** <https://huggingface.co/allenai/olmOCR-2-7B-1025> (primary)

****Supporting quote:**** "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset. It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:23:22.593Z)

I'll verify this claim about olmOCR-2-7B. Let me check the primary source and search for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:23:25.438Z)

Now let me fetch the primary source and search for corroborating evidence in parallel.

4. user (2026-05-31T04:23:32.720Z)

olmOCR-2-7B-1025 Model Details

Base Model

This model is fine-tuned from **Qwen2.5-VL-7B-Instruct**.

Training Methods

The model description explicitly states:

```
> "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the [olmOCR-mix-1025](https://huggingface.co/datasets/allenai/olmOCR-mix-1025) dataset. It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases".
```

Specific Capabilities Targeted

The model targets:

- **Math equations**
- **Tables**
- **Other tricky OCR cases**

License

The model is licensed under **Apache 2.0**:

```
> "This model is licensed under Apache 2.0. It is intended for research and educational use in accordance with Ai2's [Responsible Use Guidelines](https://allenai.org/responsible-use)."
```

The base model (Qwen2.5-VL-7B-Instruct) would have its own licensing terms on its model card, though specific details aren't provided in this content excerpt.

Key Training Datasets

- **SFT Dataset**: [olmOCR-mix-1025](https://huggingface.co/datasets/allenai/olmOCR-mix-1025)
- **RL Dataset**: [olmOCR-synthmix-1025](https://huggingface.co/datasets/allenai/olmOCR-synthmix-1025)

5. user (2026-05-31T04:23:35.030Z)

Web search results for query: "olmOCR-2-7B-1025 Qwen2.5-VL-7B GRPO RL training math equations tables"

Links: [{"title": "olmOCR-2-7B ? Doubleword | Document OCR Model", "url": "https://www.doubleword.ai/models/olmocr-2-7b/"}, {"title": "allenai/olmOCR-2-7B-1025 · Abrazando la cara", "url": "https://translate.google.com/translate?u=https%3A%2F%2Fhuggingface.co%2Fallenai%2FolmOCR-2-7B-1025&hl=es&sl=en&tl=es&client=srp"}, {"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}, {"title": "Mungert/olmOCR-2-7B-1025-GGUF · Hugging Face", "url": "https://huggingface.co/Mungert/olmOCR-2-7B-1025-GGUF"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "olmOCR-2-7B-1025-FP8 huggingface.co api & allenai olmOCR-2-7B-1025-FP8 github AI Model - Toolify", "url": "https://www.toolify.ai/ai-model/allenai-olmocr-2-7b-1025-fp8"}, {"title": "olmOCR 2 Unit Test Rewards for Document OCR", "url": "https://arxiv.org/html/2510.19817v1"}, {"title": "(PDF) olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"}, {"title": "richardyoung/olmocr2", "url": "https://ollama.com/richardyoung/olmocr2"}, {"title": "Can GRPO Boost Complex Multimodal Table Understanding?", "url": "https://arxiv.org/pdf/2509.16889"}]

Based on the search results, here's what I found about olmOCR-2-7B-1025 and its training approach:

Overview

olmOCR-2-7B-1025 is a release of the olmOCR model fine-tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset.

Training Method

The model has been additionally fine-tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases. More specifically, olmOCR 2 is powered by olmOCR-2-7B-1025, a specialized 7B vision language model trained using reinforcement learning with verifiable rewards (RLVR), where rewards are a diverse set of binary unit tests.

Key Results

RL training on test cases results in state-of-the-art performance on olmOCR-Bench, with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions.

Output Format

The model outputs natural-reading plain text with LaTeX for equations and HTML for tables.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:23:38.423Z)

The claim is strongly supported by the primary source. The arXiv paper "olmOCR 2: Unit Test Rewards for Document OCR" (arxiv.org/html/2510.19817v1) corroborates it. Let me verify the GRPO terminology specifically, since the claim uses "GRPO" while the paper abstract mentions "RLVR" ? I want to confirm these aren't being conflated and that "GRPO" is accurate.

7. user (2026-05-31T04:23:43.930Z)

olmOCR 2: GRPO and Training Details

****Does the paper use GRPO?****

Yes. The paper states: "We apply Group Relative Policy Optimization (GRPO) (Shao et al., [2024]) to olmOCR using our synthetic verifiable unit tests as binary-valued reward signals."

****What is the base model?****

The foundation is Qwen 2.5 VL. According to the paper: "We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

****Does RL training target math and tables specifically?****

Not exclusively. While the abstract notes "the largest improvements in math formula conversion, table parsing, and multi-column layouts," the RL training uses comprehensive unit tests. The paper emphasizes broad applicability: the reward system checks text presence, reading order, table accuracy, math formulas, and baseline robustness across diverse document types, rather than targeting these categories in isolation.

8. assistant (2026-05-31T04:23:53.184Z)

All key claim components are verified against both the primary HuggingFace source and the peer-facing arXiv paper:

1. *****document-OCR-specialist VLM***** ? confirmed; it's a specialized document OCR model.
2. *****fine-tuned from Qwen2.5-VL-7B-Instruct***** ? confirmed verbatim by both sources.
3. *****additional GRPO RL training***** ? confirmed. The arXiv paper explicitly states "We apply Group Relative Policy Optimization (GRPO) (Shao et al., 2024)." The HF card says "GRPO RL training." The claim's "GRPO" terminology is accurate, not a conflation with the broader "RLVR" framing.
4. *****specifically to improve performance on math equations and tables***** ? confirmed. HF card: "to boost its performance at math equations, tables, and other tricky OCR cases." The arXiv

abstract notes "largest improvements in math formula conversion, table parsing, and multi-column layouts."

The only minor nuance: the claim says GRPO was added "specifically" for math/tables. The arXiv paper clarifies the RL uses comprehensive unit tests across many document aspects (not math/tables in isolation), but math equations and tables were where the largest gains landed and are explicitly named targets. This is a slight emphasis difference, not a misrepresentation ? the claim mirrors the HF card's own wording.

The claim's final editorial gloss ("the harder cases for bank-statement parsing") is the claimant's own framing for the muneem context, not attributed to AllenAI, and is reasonable. Source quality is high (primary model card + arXiv technical report). Not outdated (1025 = Oct 2025 release). Not marketing fluff ? corroborated by the technical paper.

The claim survives all three refutation attempts.

9. user (2026-05-31T04:24:00.840Z)

Structured output provided successfully

10. assistant (2026-05-31T04:24:04.198Z)

Claim verified as ****not refuted**** with high confidence. Both the primary HuggingFace model card and the peer-facing arXiv technical paper (2510.19817v1) independently confirm every component: olmOCR-2-7B is a document-OCR-specialist VLM fine-tuned from Qwen2.5-VL-7B-Instruct, with additional GRPO (Group Relative Policy Optimization) RL training that targets math equations and tables. The arXiv paper confirms the GRPO algorithm name verbatim, ruling out any conflation with the broader RLVR framing.